Data Visualization GUI Interactivity Study

Booking (Ahead of time)

- Send consent form

Checklist (Preparation Before Study)

A. 3 Copies Consent Form
B. Charge any devices needed
C. 2 Pens
D. Verify with each of the **X** participants:
    - Time and place of meeting
    - Contact info:
        - Phone number
        - Address
    - Who will be there
    - Go over any consent questions if lingering
E. Surveys:
    - 2 background questionnaires

Setup

- Set up design materials
- Assign the participant an anonymous ID

Consent (~10min)

Introduction:
Welcome! Thank you for participating in this study. Throughout the study, I may need to read from a script to ensure that I keep our session on track.

The purpose of this study is to investigate user interactions with dashboard widgets, graphs, the carousel, sidebars, and the overall data visualization pipeline in a web-based AI-driven platform. By studying how participants engage with these features, we aim to improve their design and usability.

This study is being conducted by myself, George Morales.

Important Information

Many of the study tasks involve answering questions about your background and experience with data visualization tools. You will also be asked to interact with various dashboard elements, such as uploading datasets, labeling columns, generating graphs, customizing widgets, and navigating the carousel. Additionally, we will conduct a semi-structured interview to discuss your experience, challenges, and preferences.

Your participation is entirely voluntary, and you may choose to withdraw at any time. If you have any questions before we begin, feel free to ask.

Study Procedure

Today's session consists of two main activities.
First, I will ask you to upload a dataset, describe it, and label its columns for visualization.

Next, I will introduce you to the dashboard workspace, where you can interact with data visualizations, graphs, and charts. You will have the opportunity to explore the interface, adjust settings, and experiment with different configurations to see how the system responds.

Then, I will ask you to create and customize dashboard widgets, rearrange elements within the carousel, and interact with the sidebar to refine the display of your visualizations.

Finally, I will ask you to provide feedback on your experience, share any challenges or preferences you encountered, and discuss potential improvements to the interaction design.

Design Presentation (~5-10min.)

This will be done on my laptop. The browser-based GUI design consists of two main parts: data upload and annotation and dashboard interactions. Users start by uploading a dataset (CSV/Excel), labeling columns, and adding descriptions for context. Once processed, they enter the interactive dashboard, where they can generate and customize graphs, explore widgets, and navigate visualizations within a carousel. Sidebars provide additional options for filtering and refining insights.

Interview (~15min.)

I'd like to ask you a few questions about your experience interacting with the dashboard, including the data upload process, annotation, and visualization features. Please share your thoughts as openly as possible.

1. ***How intuitive was the data upload and annotation process? Were there any challenges or areas for improvement?***

   Uploading the dataset was straightforward, and the annotation process felt well-structured. The system did a good job of guiding users through labeling and describing columns. One minor frustration was that navigating between columns required extra clicks, which slowed things down. A more seamless way to move between columns, like a next/previous shortcut or bulk editing mode, could be convenient.

2. ***How easy was it to navigate and interact with the dashboard, widgets, graphs, and carousel?***

   The dashboard layout was intuitive, and interacting with widgets and graphs felt natural. The carousel made it easy to browse different visualizations, but it took up space even when not in use. Having an option to collapse or minimize it would make the interface feel less cluttered. Also, while the widgets were functional, being able to resize them freely would improve the ana analysis of data points.

3. ***Were the customization options sufficient for refining visualizations? If not, what additional features would you find useful?***

   The existing customization options covered the basics well, and the ability to expand widgets and hover over data points was useful. However, the lack of widget resizing stood out—being able to adjust widget dimensions freely would be a big game changer. It would allow for better layout control, especially when comparing multiple visualizations at once. Adding custom grid layouts or snapping widgets into place could improve usability.

4. ***Did anything about the interface feel confusing, frustrating, or particularly helpful? Why?***

   The interface was generally easy to use, and the guided workflow made setting up visualizations simple. However, one major frustration was that widgets couldn't be moved around freely—they were locked in place. This made it harder to organize the dashboard in a way that suited my own creative needs.

[My Notes]

- *Dataset upload and annotation are clear, but navigating between columns could be smoother.*
- *Manually selecting data types is tedious; bulk editing or auto-detection would help.*
- *Widgets are static. Resizing them would be a major improvement.*
- *Widgets being locked in place is frustrating—drag-and-drop functionality is needed for better customization.*

Conclusion

Thank you for your help with our study. We are finished with all the session's tasks.